Northern Berkshire Astronomical Society

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This Month

Mercury, Jupiter, and the Moon

The Moon

- Feb 1: Wolf Moon

- Feb 9

- Feb 17

- Feb 24

Planets

Mercury: behind the Sun

Venus: behind the Sun

Mars: behind the Sun

Jupiter: in Gem, up all night

Saturn: in Psc, sets ~8 PM

Uranus: in Tau, up all evening

Neptune: in Psc, sets ~8 PM

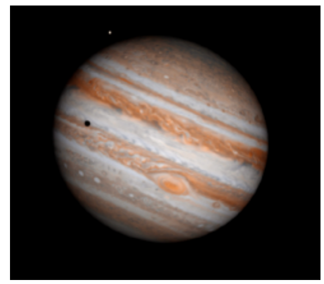
Deep Sky Objects

Easy (binoculars): M 42/43 (Ori), M 41 (CMa), M 36, 37, and 38 (Aur), M 46, 47, and 93 (Pup)

Moderate (small telescopes): M 79 (Lep), NGC 2264

Challenges: B 22, C 49/50, B 33 (Horsehead), NGC 2024

Catching Jupiter's Great Red Spot



With Jupiter high in the sky in Gemini, and close to opposition (back on Jan 10), this is a good time attempt some "close-up" observing of the great planet.

Jupiter's rotation period is about 9h 55m, though the different bands and zones have slightly different values. The most famous feature is the Great Red Spot (GRS). When it first was observed is a little unclear: definite observations go back to 1831, though mentions of a Jupiter "spot" go back to 1664, though some of them might've been shadow transits of the Galilean moons, or a different "spot" altogether.

The spot we see now is an enormous cyclone whose top extends past the bands and zones crossing the planet. It's about the size of the Earth though it has shrunk over the last decades, and – sadly – dimmed but it's still within reach of moderate telescopes.

For February, here are the transit times (EST) during evening hours. The "window of opportunity" is ~45 min on either side of these times:

Date/EST	Date/EST	Date/EST
Feb 5: 6:31 PM	Feb. 14: 8:55 PM	Feb 24: 7:12 PM
Feb 7: 8:09 PM	Feb 16: 10:34 PM	Feb 26: 8:50 PM
Feb 9: 9:47 PM	Feb 19: 8:03 PM	Mar 3: 7:58 PM
Feb 11: 11:26 PM	Feb 21: 9:42 PM	

This Month's Image

Messier 46 in Puppis is a two-fer!

Cataloged as an open cluster, it's typically overshadowed by other more-famous winter objects, but it's an easy find in NW Puppis (and next to neighboring cluster M 47). It's bright enough to be a easy target for small telescopes (even binoculars), but the cluster (5000 ly away) is being "photobombed" by a closer (1370 ly) planetary nebula that started forming 8.5 kyr ago when its parent star shed its outer atmosphere leaving a hot core heating the shed gases and creating the smoky ring of the nebula. As it expands and the stellar remnant cools, it will slowly fade away.

Interacting

Check out our Facebook Group

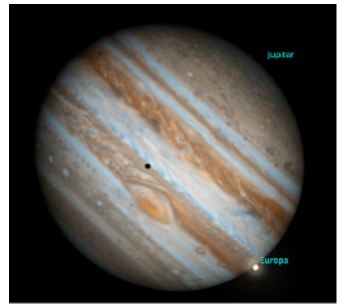
https://www.facebook.com/groups/nberkastro

and join us at our next meeting: **March 4th at 6 PM** at the North Adams Public Library

Or the GRS and a Moon Shadow

If you can brave the cold into the late evening/early morning there are also opportunities to see the Great Red Spot along with a shadow transit of one of Jupiter's moons! Here are four such opportunities this month:

Date / EST	Moon
Feb 4: 11:15 PM	Io
Feb 16: 3:00 AM	Europa
Feb 18: 11:35 PM	Ganymede
Feb 26: 1:24 AM	Ganymede



For these the GRS might not be on the meridian, but it'll be close to it. The Europa transit on the 16th will have the sunlit moon just on of the edge of Jupiter's disk with the shadow near the center.

Mercury in the Evening

2026 isn't the best year for catching Mercury, but the best chance might come on the evenings surrounding Feb 19th and 20th. While only 15° from the Sun, it sets right at the end of twilight, around 7 PM (the Sun sets at 5:30). The thin crescent Moon might help: it'll be higher in the sky on the night of the 20th and "points" to Mercury's position. (The night before the Moon will be just below the planet but with a *very* thin crescent!) Saturn is there too, but will be a magnitude fainter than Mercury.